

Total No. of Questions : 8]

SEAT No. :

PD-4503

[Total No. of Pages : 2

[6403]-312

TE (Honours in Data Science)
DATA SCIENCE AND VISUALIZATION
(2019 Pattern) (Semester - V) (310501)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidate:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q. 5 or Q. 6, Q.7, or Q. 8,
- 2) Neat diagrams must be drawn wherever necessary
- 3) Figures to the right indicate full marks
- 4) Assume suitable data, if necessary

- Q1)** a) Describe the concept of clustering and discuss the challenges associated with clustering in large datasets [6]
- b) What are the steps involved in building a linear regression model? Explain with an example. [6]
- c) Explain the concept of interestingness measures in association rule mining. Why are they important? [6]

OR

- Q2)** a) Describe how Naïve Bayes classification can be used for text categorization tasks [6]
- b) Differentiate between hierarchical clustering and Distribution model- based clustering methods [6]
- c) Write a note on association rules and its applicability. [6]
- Q3)** a) Explain the process of building a Decision Tree using the ID3 algorithm [9]
- b) What is a Feed-Forward Neural Network? Explain its architecture with a neat diagram. [8]

OR

P.T.O.

- Q4)** a) What is MapReduce? Explain the role of MapReduce in handling data with examples of Word Count and Matrix Multiplication. [9]
- b) What is Backpropagation? Describe its working with an example. [8]

- Q5)** a) Discuss the key challenges faced during data visualization and suggest ways to overcome them. [9]
- b) Explain different types of dashboards (strategic, analytical, operational) with examples [9]

OR

- Q6)** a) Write a note on: [6]
- i) Bar Charts
- ii) Line Charts
- iii) Area Charts
- b) What are statistical visualizations? Explain with suitable examples. [6]
- c) Explain the principles for designing an effective dashboard. [6]

- Q7)** a) What is the need for visualizing multidimensional data? Explain with suitable cases. [6]
- b) Explain Entity - Relationship (ER) Data Modelling. [6]
- c) What are the steps in performing Principal Component Analysis (PCA) on a dataset? [5]

OR

- Q8)** a) Differentiate between logical data models and physical data models. [6]
- b) What are the key steps in data modelling process? Enlist and Explain [6]
- c) What are the major challenges when performing clustering on high-dimensional data? Suggest some solutions [5]

